

Predicted

Actual

TP	FN
FP	TN

F P (predicted)
N (real)

$$P = \frac{TP}{TP+FP}$$

	Predicted	
Actual	TP	FN
	FP	TN

Hamro model le Positive predict gareko madhye kati chai sachikai Positive thiye?

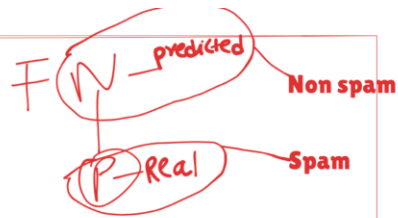
$$R = \frac{TP}{TP+FN}$$

	Predicted	
Actual	TP	FN
	FP	TN

Out of all the actual positives, how many were correctly identified as positive by our model?

Mail
 ↳ SPAM (Positive)
 ↳ nonSPAM (Negative)

P — reduces FP
 R — reduces FN



FP means (Real ma neg thiyo
 tara pos classify vayo)
 $\therefore FP \rightarrow \frac{\text{nonSPAM} \xrightarrow{\text{truth}} \text{SPAM} \xrightarrow{\text{predict}} \text{SPAM}}$
 nonSPAM lai SPAM pred
 garna ghatana precision
 use garna paryo

FN means (Real ma positive thiyo
 tara classify garda neg vayo)
 $\therefore FN \rightarrow \frac{\text{SPAM} \xrightarrow{\text{truth}} \text{SPAM} \xrightarrow{\text{predict}} \text{Non spam}}$
 SPAM lai non spam
 predict garna ghatana
 recall use garna paryo

what is the major problem in our case ?

- non-spam lai ne spam vandibo vane ta maro / mP
 mail s ilre (interview promotion) miss hoina
 - SPAM lai nonSPAM vane bary afai mail harna delete
 garna saka
- $\therefore$ so nonSPAM — SPAM dangerous
 i.e FP is dangerous to reduce
 we should use precision (High)

Disease (Positive)
 Cancer (Negative)
 P - reduces FP
 R - reduces FN

FP means (Real ma neg thing but +ve vaju)

$\therefore FP$ (non cancer) ———— (cancer)

(non cancer patient lai cancer predict garna ghadanna FP lai impure garna parjo)

FN means (Real ma +ve thing but -ve vaju)

$\therefore FN$ (cancer) ———— (non cancer)

(cancer lai non cancer predict garna ghadanna FN lai impure garna parjo)

what is the major problem in our case ?

- non cancer lai cancer vanda baro problem hudaina kina vane person can further do other tests and wont die.
- cancer lai non cancer vanda te problem vaju heta she can die so this is the major problem (FN) so we should reduce it
 So we should improve FN

precision	recall	f1-score	support
0.94	1.00	0.97	15

Hamro model le Obese bhanera predict gareka manxeharu madhye 94% caess ma sachikai Obese rahechan.

Out of all actual obesity cases, the model correctly identified every single one as obesity.
Out of all actual obesity cases, the model correctly identified 70% as obesity.

15 ota actual obesity cases madhye model le 15 ota nai obesity bhanera sahi chinya. Euta pani obesity case miss bhayena no actual positive cases were missed.

